Lathi 20题练习册 第7批

对应 Boulet 前四章

原书英文题干、公式与必要图示 · 无答案版

记号：D = d/dt；差分式中 E 为前移算子，E{y[n]} = y[n+1]；j^2 = -1
u 为单位阶跃，δ 为单位冲激；LTIC / LTID 分别指连续 / 离散时间 LTI 系统

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来源：B. P. Lathi / R. A. Green, Linear Systems and Signals, 3rd ed. (2018)
中文主题仅用于课程对应；题目要求以原书英文为准。

正交相位方波及复信号的能量与功率

Lathi 1.1-8 | 原书第 137 页

Boulet 第1章：Periodic Signals, p.8；Finite-Energy and Finite-Power Signals, p.21

原题同页跨栏，两个信号的定义及(a)-(d)完整附上。

1.1-8 Two periodic signals that differ only by a 90-degree phase shift are considered to be quadrature signals. For example, $\cos(2\pi t)$ and $\sin(2\pi t)$ are quadrature signals. Another pair of quadrature signals is $x(t) = \text{sgn}[\cos(2\pi t)]$ and $y(t) = \text{sgn}[\sin(2\pi t)]$, where sgn is the sign (or signum) function.

(a) Plot $x(t)$ and determine its power P_x and energy E_x.

(b) Plot $y(t)$ and determine its power P_y and energy E_y.

(c) Consider the complex function $f(t) = x(t) + jy(t)$. Determine the power and energy of $f(t)$.

(d) When real functions $x(t)$ and $y(t)$ are combined as $f(t) = x(t) + jy(t)$, is it generally true that $E_f = E_x + E_y$ and $P_f = P_x + P_y$? Prove your answer.

演算区

连续波形的反转、平移与时间伸缩

Lathi 1.2-1 | 原书第 138、139 页

Boulet 第1章：Transformations of the Time Variable, p.4

1.2-1 For the signal $x(t)$ depicted in Fig. P1.2-1, sketch the signals

(a) $x(-t)$

(b) $x(t+6)$

(c) $x(3t)$

(d) $x(t/2)$

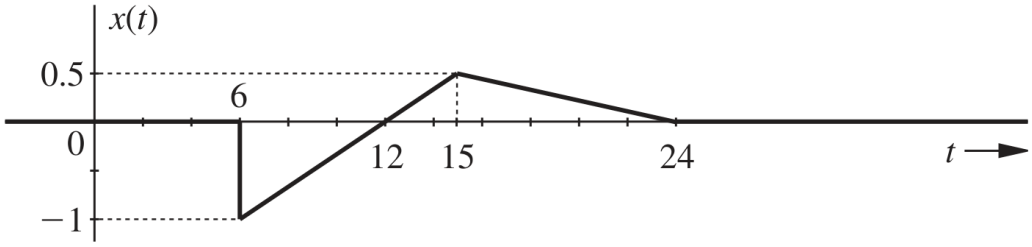


Figure P1.2-1

演算区

奇偶分解后的时间变换

Lathi 1.5-2 | 原书第 142、143 页

Boulet 第1章：Even and Odd Signals, p.23；Transformations of the Time Variable, p.4

原题跨页，(a)、(b)均已附。

1.5-2 Define $x(t) = 2u(t+1) - u(t-2) - u(t-3)$.

(a) Letting $x_o(t)$ designate the odd portion of $x(t)$, accurately sketch $x_o(1-2t)$.

(b) Letting $x_e(t)$ designate the even portion of $x(t)$, accurately sketch $x_e(2+t/3)$.

演算区

延迟半波整流系统的基本性质

Lathi 1.7-5 | 原书第 145 页

Boulet 第1章：System Models and Basic Properties，p.34

展开所引用1.7-4的要求：分别判断系统是否(a)BIBO稳定、(b)因果、(c)可逆、(d)线性、(e)无记忆、(f)时不变；每项回答是或否，并给出数学论证。

1.7-5 Repeat Prob. 1.7-4 for a system with input $x(t)$ that produces output $y(t)$ according to

$$y(t+1) = \begin{cases} -2x(t) & \text{when } x(t) \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

演算区

用单位阶跃构造有限离散序列

Lathi 3.3-3 | 原书第 316 页

Boulet 第1章：Discrete-Time Impulse and Step Signals, p.25

3.3-3 Sketch the following signals:

(a) $u[n-2]-u[n-6]$

(b) $n\{u[n]-u[n-7]\}$

(c) $(n-2)\{u[n-2]-u[n-6]\}$

(d) $(-n+8)\{u[n-6]-u[n-9]\}$

(e) $(n-2)\{u[n-2]-u[n-6]\}+(-n+8)\{u[n-6]-u[n-9]\}$

演算区

正弦窗与分段常数信号卷积的极小值位置

Lathi 2.4-3 | 原书第 225 页

Boulet 第2章：Continuous-Time LTI Systems: The Convolution Integral, p.67

2.4-3 Consider signals $h(t) = -u(t+2) + 3u(t-1) - 2u(t-\frac{5}{2})$ and $x(t) = \sin(t)\,[u(t+2\pi) - u(t+\pi)]$. Determine the approximate time $t_{\min}$ where $y(t) = x(t) * h(t)$ is a minimum. Note, the minimum value of $y(t) \neq 0$!

演算区

由信号图直接判断卷积值并求完整响应

Lathi 2.4-22 | 原书第 227、228 页

Boulet 第2章：Continuous-Time LTI Systems: The Convolution Integral，p.67

原题跨页，(a)、(b)及输入和冲激响应双图均已附。

2.4-22 Figure P2.4-22 shows the input $x(t)$ and the impulse response $h(t)$ for an LTIC system. Let the output be $y(t)$.

(a) By inspection of $x(t)$ and $h(t)$, find $y(-1), y(0), y(1), y(2), y(3), y(4), y(5)$, and $y(6)$. Thus, by merely examining $x(t)$ and $h(t)$, you are required to see what the result of convolution yields at $t = -1, 0, 1, 2, 3, 4, 5$, and 6.

(b) Find the system response to the input $x(t)$.

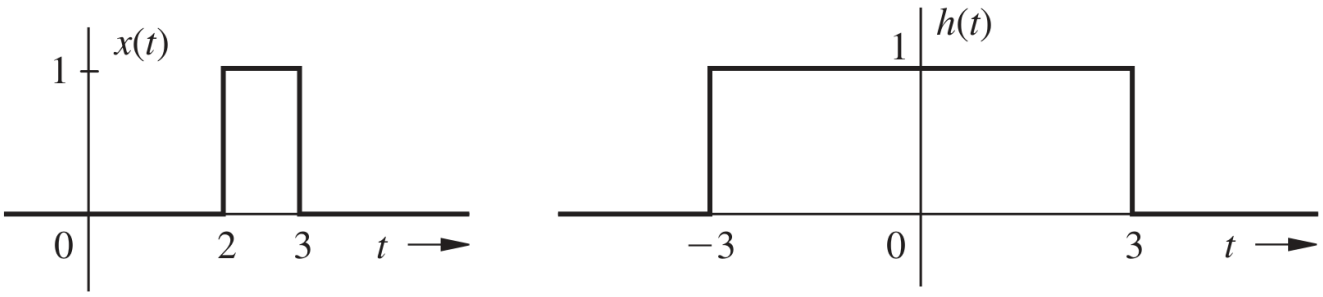


Figure P2.4-22

演算区

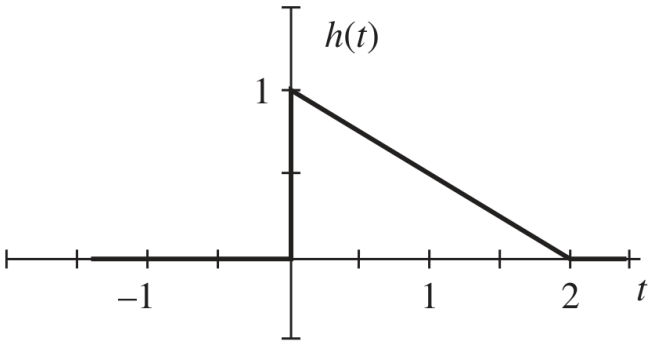
两个连续LTI系统串联的指定时刻阶跃响应

Lathi 2.4-31 | 原书第 228、230 页

Boulet 第2章：Properties of Linear Time-Invariant Systems，p.74；Continuous-Time LTI Systems: The Convolution Integral，p.67

所需原图 P2.4-31 的 h(t) 波形和串联框图均已附。

2.4-31 Two linear time-invariant systems, each with impulse response $h(t)$, are connected in cascade. Refer to Fig. P2.4-31. Given input $x(t) = u(t)$, determine $y(1)$. That is, determine the step response at time $t = 1$ for the cascaded system shown.



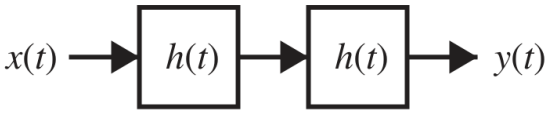


Figure P2.4-31

演算区

周期截断离散冲激响应的卷积与特征根

Lathi 3.8-3 | 原书第 321 页

Boulet 第2章：Discrete-Time LTI Systems: The Convolution Sum，p.54；兼及第3章 Characteristic Polynomials and Stability，p.112

3.8-3 Let $x[n] = (0.5)^n\,(u[n+4] - u[n-4])$ be input into an LTID system with an impulse response given by

$$h[n] = \begin{cases} 2 & [(n \bmod 6) < 4] \text{ and } [n \geq 0] \\ 0 & \text{otherwise} \end{cases}$$

Recall, $(n \bmod p)$ is the remainder of the division n/p. The system is described according to the difference equation $y[n] - y[n-6] = 2x[n] + 2x[n-1] + 2x[n-2] + 2x[n-3]$.

(a) Determine the six characteristic roots (γ_1 through γ_6) of the system.

(b) Determine the value of $y[10]$, the zero-state output of system $h[n]$ in response to $x[n]$ at time $n = 10$. Express your result in decimal form to at least three decimal places (e.g., $y[10] = 3.142$).

演算区

单位阶跃响应与冲激响应的累积关系

Lathi 3.8-19 | 原书第 323、296 页

Boulet 第2章：Discrete-Time LTI Systems: The Convolution Sum，p.54；Properties of Linear Time-Invariant Systems，p.74

所引用式(3.37)已附。其中 h[n] 是单位冲激响应，g[n] 是单位阶跃响应。

3.8-19 (a) Show that for a causal system, Eq. (3.37) can also be expressed as

$$g[n] = \sum_{k=0}^{n} h[n-k]$$

(b) How would the expressions in part (a) change if the system is not causal?

$$g[n] = \sum_{k=-\infty}^{n} h[k] \qquad (3.37)$$

演算区

纯虚特征根连续系统的初值响应

Lathi 2.2-7 | 原书第 224 页

Boulet 第3章：Causal LTI Systems Described by Differential Equations，p.92；Characteristic Polynomials and Stability，p.112

展开所引用2.2-4的要求：(a)求特征多项式、特征方程、特征根和特征模态；(b)按给定初值求 t>=0 的零输入响应。

2.2-7 Repeat Prob. 2.2-4 for

$$(D^2+9)y(t)=(3D+2)x(t)$$

and $y_0(0^-)=0, \dot{y}_0(0^-)=6.$

演算区

运放RC电路的微分方程与储能初值响应

Lathi 2.2-12 | 原书第 224 页

Boulet 第3章：Causal LTI Systems Described by Differential Equations，p.92

题干、元件符号与极性完整保留；本题先建模，再按给定初始电容电压求零输入响应。

2.2-12 Consider the circuit of Fig. P2.2-12. Using operator notation, this system can be described as $(D+a_1)\{y(t)\} = (b_0D+b_1)\{x(t)\}$.

(a) Determine the constants a_1, b_0, and b_1 in terms of the system components R, R_f, and C.

(b) Assume that $R = 300\ \text{k}\Omega$, $R_f = 1.2\ \text{M}\Omega$, and $C = 5\ \mu\text{F}$. What is the zero-input response $y_0(t)$ of this system, assuming $v_C(0) = 1\ \text{V}$?

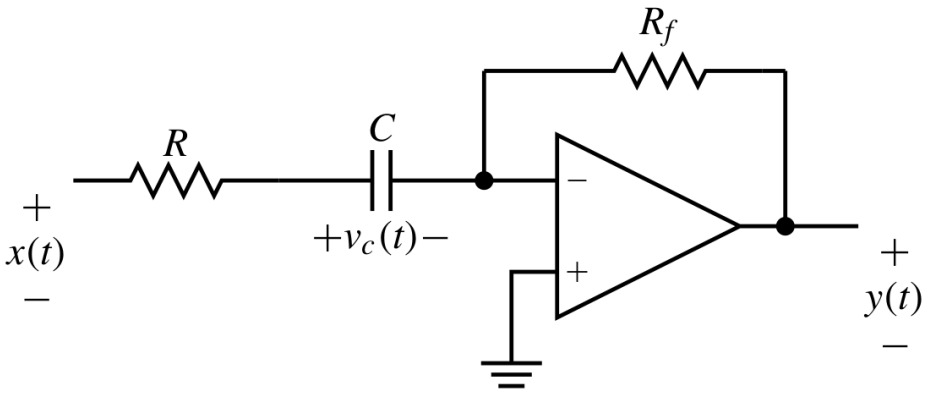


Figure P2.2-12

演算区

二阶离散系统的前三项递推响应

Lathi 3.5-6 | 原书第 320 页

Boulet 第3章：Causal LTI Systems Described by Difference Equations, p.96

展开所引用3.5-5的要求：按给定输入及初始条件递推求解，只求前三项。

3.5-6 Repeat Prob. 3.5-5 for

$$y[n]+2y[n-1]+y[n-2]=2x[n]-x[n-1]$$

$$\text{with } x[n]=(3)^{-n}u[n], y[-1]=2, \text{and } y[-2]=3.$$

演算区

重特征根齐次差分方程的初值解

Lathi 3.6-3 | 原书第 320 页

Boulet 第3章：Causal LTI Systems Described by Difference Equations， p.96

3.6-3 Solve

$$y[n+2]+2y[n+1]+y[n]=0$$

if $y[-1]=1$ and $y[-2]=1$.

演算区

一阶离散方程中下标位置与冲激响应

Lathi 3.7-1 | 原书第 320 页

Boulet 第3章：Impulse Response of a Difference LTI System， p.109

3.7-1 Find the unit impulse response $h[n]$ of systems specified by the following equations:

(a) $y[n+1]+2y[n]=x[n]$

(b) $y[n]+2y[n-1]=x[n]$

演算区

周期指数波的时间反转与幅相谱

Lathi 6.1-2 | 原书第 669、670、599、600 页

Boulet 第4章：求傅里叶级数，p.134；线谱，p.137；Time Reversal，p.140

题干(a)至(c)、原图P6.1-2、引用的图6.2(a)及例6.1已知级数均已附。参考信号x(t)在0<t<π时为exp(-t/2)，以π为周期延拓，基频为2 rad/s。参考级数中的0.504是2(1-exp(-π/2))/π的四舍五入值，精确计算可保留该表达式。所附参考级数是(b)的已知条件，不是本题y(t)的答案。

6.1-2 (a) Find the trigonometric Fourier series for $y(t)$ shown in Fig. P6.1-2.

(b) The signal $y(t)$ can be obtained by time reversal of $x(t)$ shown in Fig. 6.2a. Use this fact to obtain the Fourier series for $y(t)$ from the results in Ex. 6.1. Verify that the Fourier series thus obtained is identical to that found in part (a).

(c) Show that, in general, time reversal of a periodic signal does not affect the amplitude spectrum, and the phase spectrum is also unchanged except for the change of sign.

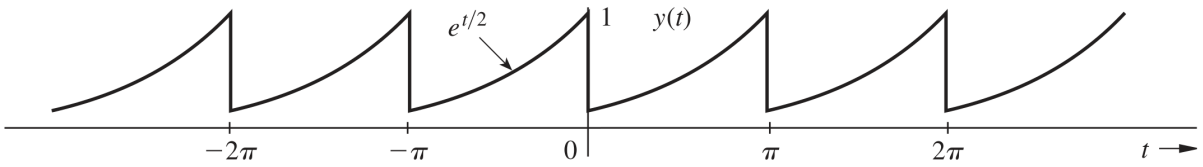
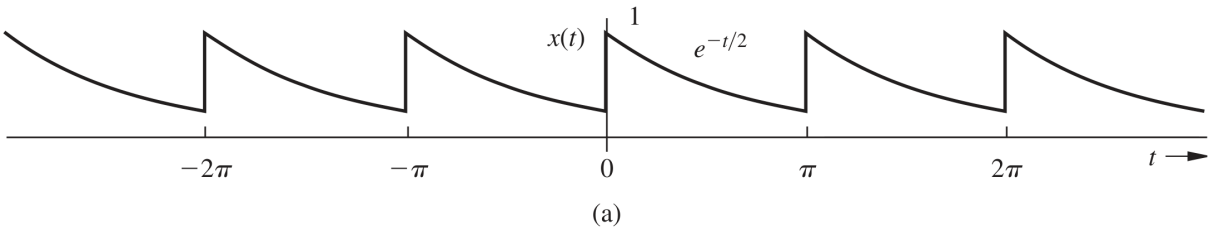


Figure P6.1-2



(a)

$$x(t) = 0.504\left[1 + \sum_{n=1}^{\infty} \frac{2}{1+16n^2}(\cos 2nt + 4n \sin 2nt)\right]$$

周期指数波的时间压缩与频谱展宽

Lathi 6.1-3 | 原书第 669、671、599、600 页

Boulet 第4章：求傅里叶级数，p.134；线谱，p.137；Time Scaling，p.140

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6.1-3 (a) Find the trigonometric Fourier series for the periodic signal $y(t)$ depicted in Fig. P6.1-3.

(b) The signal $y(t)$ can be obtained by time compression of $x(t)$ shown in Fig. 6.2a by a factor 2. Use this fact to obtain the Fourier series for $y(t)$ from the results in Ex. 6.1. Verify that the Fourier series thus obtained is identical to that found in part (a).

(c) Show that, in general, time compression of a periodic signal by a factor a expands the Fourier spectra along the ω axis by the same factor a. In other words C_0, C_n, and θ_n remain unchanged, but the fundamental frequency is increased by the factor a, thus expanding the spectrum. Similarly, time expansion of a periodic signal by a factor a compresses its Fourier spectra along the ω axis by the factor a.

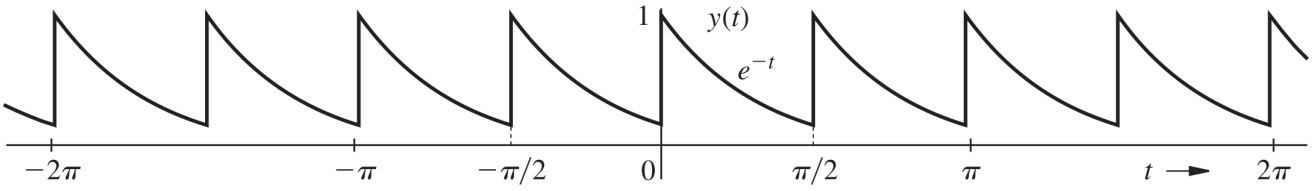


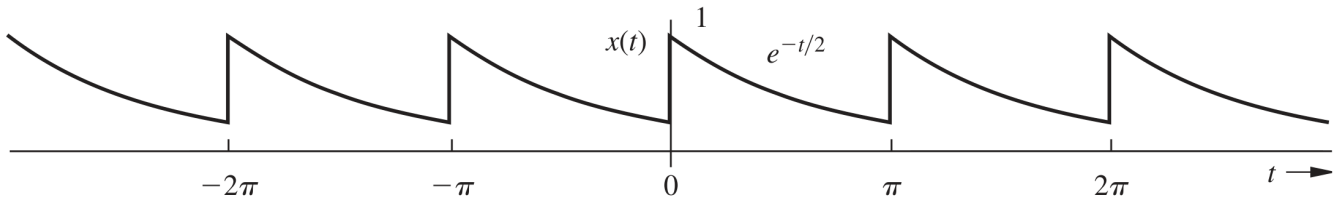
Figure P6.1-3

周期指数波的时间压缩与频谱展宽（续）

Lathi 6.1-3 | 原书第 669、671、599、600 页

Boulet 第4章：求傅里叶级数，p.134；线谱，p.137；Time Scaling，p.140

题干(a)至(c)、原图P6.1-3、引用的图6.2(a)及例6.1已知级数均已附。参考信号x(t)在0<t<π时为exp(-t/2)，以π为周期延拓，基频为2 rad/s；题图y(t)在0<t<π/2时为exp(-t)，以π/2为周期延拓。参考级数中的0.504是2(1-exp(-π/2))/π的四舍五入值。所附参考级数是(b)的已知条件，不是本题y(t)的答案。



(a)

$$x(t) = 0.504\left[1 + \sum_{n=1}^{\infty} \frac{2}{1+16n^2}(\cos 2nt + 4n \sin 2nt)\right]$$

演算区

三角波的时移、偶对称性与傅里叶相位

Lathi 6.1-4 | 原书第 669、671、602、603 页

Boulet 第4章：求傅里叶级数，p.134；Time Shifting，p.139；对称性，pp.140-141

原题跨栏，(a)至(c)均已附；另附P6.1-4、图6.4(a)和例6.2已知级数(6.12)。参考信号x(t)的周期为2：在-1/2<t<1/2时x(t)=2At，在1/2<t<3/2时x(t)=2A(1-t)，再作周期延拓；A为图示幅值。已知级数只用于(b)的时移推导，未附g(t)的解答。

6.1-4 (a) Find the trigonometric Fourier series for the periodic signal $g(t)$ in Fig. P6.1-4. Take advantage of the symmetry.

(b) Observe that $g(t)$ is identical to $x(t)$ in Fig. 6.4a left-shifted by 0.5 second. Use this fact to obtain the Fourier series for $g(t)$ from the results in Ex. 6.2. Verify that the Fourier series thus obtained is identical to that found in part (a).

(c) Show that, in general, a time shift of T seconds of a periodic signal does not affect the amplitude spectrum. However, the phase of the nth harmonic is increased or decreased $n\omega_0 T$ depending on whether the signal is advanced or delayed by T seconds.

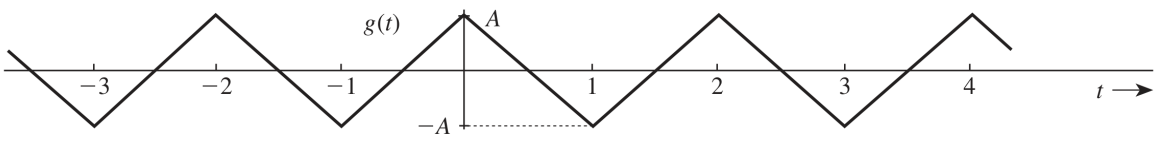
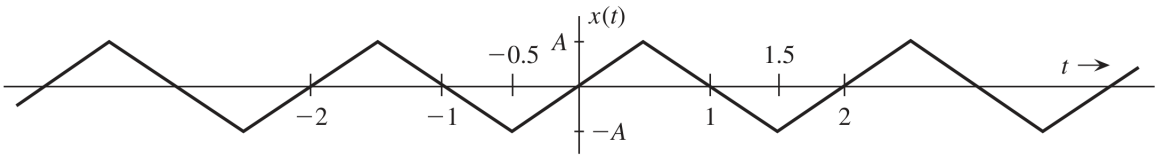


Figure P6.1-4



(a)

$$x(t) = \frac{8A}{\pi^2}\left[\sin \pi t - \frac{1}{9}\sin 3\pi t + \frac{1}{25}\sin 5\pi t - \frac{1}{49}\sin 7\pi t + \cdots\right] \qquad (6.12)$$

周期抛物线波的Parseval验证与截断误差功率

Lathi 6.3-11 | 原书第 674、609、608 页

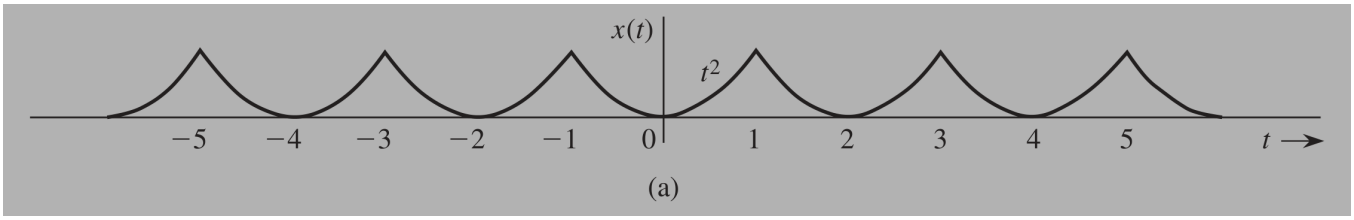
Boulet 第4章：Optimality and Convergence of the Fourier Series，p.144；Parseval Theorem，p.150

完整题干(a)(b)、图6.7(a)及Drill6.1(a)给出的级数均已附。图示信号在-1<t<1时x(t)=t^2，周期为2；其已知级数为1/3加上所附余弦和。所附级数是本题要求使用的已知条件，不含Parseval验证或所求截断项数的答案。原题以first N terms表述截断项数，作答时请说明是否把直流项计作一项。

6.3-11 (a) The Fourier series for the periodic signal in Fig. 6.7a is given in Drill 6.1. Verify Parseval's theorem for this series, given that

$$\sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}$$

(b) If $x(t)$ is approximated by the first N terms in this series, find N so that the power of the error signal is less than 1% of P_x.



$$\textbf{(a)}\ x(t) = \frac{1}{3} - \frac{4}{\pi^2}\left(\cos \pi t - \frac{1}{4}\cos 2\pi t + \frac{1}{9}\cos 3\pi t - \frac{1}{16}\cos 4\pi t + \cdots\right)$$
$$= \frac{1}{3} + \frac{4}{\pi^2}\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}\cos n\pi t$$

演算区

周期锯齿波的Parseval验证与谐波截断误差

Lathi 6.3-12 | 原书第 674、675、609、608 页

Boulet 第4章：Optimality and Convergence of the Fourier Series，p.144；Parseval Theorem，p.150

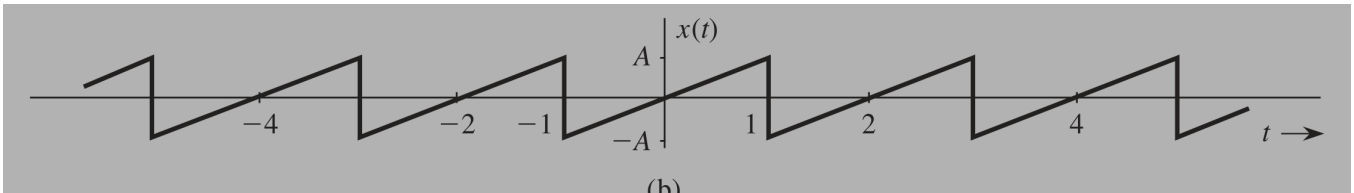
原题跨页，(a)(b)及给定的无穷级数恒等式完整附上；另附图6.7(b)及Drill6.1(b)给出的正弦式和等价紧凑三角式。图示信号在-1<t<1时x(t)=At，周期为2，A为图示幅值。所附傅里叶级数是本题的已知条件，未附功率验证或所求项数的答案。

6.3-12 (a) The Fourier series for the periodic signal in Fig. 6.7b is given in Drill 6.1. Verify

Parseval's theorem for this series, given that

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

(b) If $x(t)$ is approximated by the first N terms in this series, find N so that the power of the error signal is less than 10% of P_x.



(b)
$$x(t) = \frac{2A}{\pi}\left[\sin \pi t - \frac{1}{2}\sin 2\pi t + \frac{1}{3}\sin 3\pi t - \frac{1}{4}\sin 4\pi t + \cdots\right]$$
$$= \frac{2A}{\pi}\left[\cos(\pi t - 90^\circ) + \frac{1}{2}\cos(2\pi t + 90^\circ) + \frac{1}{3}\cos(3\pi t - 90^\circ) + \tfrac{1}{4}\cos(4\pi t + 90^\circ) + \cdots\right]$$

演算区